



Fixed Point Solutions, LLC

Hyperdrive Mint, Burn, and ERC-1155 Compatibility Assessment

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1. Scope

The Hyperdrive protocol is a DeFi primitive for fixed income trading that allows traders to go long or short interest rates, and allows liquidity providers to collect fees for providing liquidity to enable these trades. This assessment covers logic that was added to the protocol to enable new functionality:

1. Mint/Burn (<https://github.com/delvtech/hyperdrive/pull/1220>): this feature allows the simultaneous opening or closing of offsetting positions, skipping an execution of the underlying AMM logic.
2. ERC-1155 Compatibility (<https://github.com/delvtech/hyperdrive/pull/1238>): this feature adds [ERC-1155 standard](#) functionality for long and short positions of various maturities on the Hyperdrive platform.

The code was reviewed at commit [76a91bf1fe8ad32aa59d065a29037d961707c245](https://github.com/delvtech/hyperdrive/commit/76a91bf1fe8ad32aa59d065a29037d961707c245) by one reviewer for approximately 8 days. All findings were fixed by the project and all fixes were verified by the reviewer.

2. Limitations

No assessment can guarantee the absolute safety or security of a software-based system. Further, a system can become unsafe or insecure over time as it and/or its environment evolves. This assessment aimed to discover as many issues and make as many suggestions for improvement as possible within the specified timeframe. Undiscovered issues, even serious

ones, may remain. Issues may also exist in components and dependencies not included in the assessment scope.

3. Findings

Findings and recommendations are listed in this section, grouped into broad categories. It is up to the team behind the code to ultimately decide whether the items listed here qualify as issues that need to be fixed, and whether any suggested changes are worth adopting. When a response from the team regarding a finding is available, it is provided.

Findings are given a severity rating based on their likelihood of causing harm in practice and the potential magnitude of their negative impact. Severity is only a rough guideline as to the risk an issue presents, and all issues should be carefully evaluated.

Severity Level Determination		Impact		
		High	Medium	Low
Likelihood	High	Critical	High	Medium
	Medium	High	Medium	Low
	Low	Medium	Low	Low

Issues that do not present any quantifiable risk (as is common for issues in the Code Quality category) are given a severity of **Informational**.

Summary of Findings	
Severity	Count
Critical	0
High	0
Medium	0
Low	1
Informational	2

3.1 Safety and Correctness

Findings that could lead to harmful outcomes or violate the intentions of the system.

SC.1 Return Data Bombing In ERC-1155 Transfer Functions

Severity: Low

Code Location:

[1]<https://github.com/delvtech/hyperdrive/blob/76a91bf1fe8ad32aa59d065a29037d961707c245/contracts/src/internal/HyperdriveMultiToken.sol#L144>

[2]<https://github.com/delvtech/hyperdrive/blob/76a91bf1fe8ad32aa59d065a29037d961707c245/contracts/src/internal/HyperdriveMultiToken.sol#L184>

Description: The error handling for calls to `onERC1155Received()` for contract recipients of transfers will copy an arbitrary length string into memory if the called contract doesn't return the expected `bytes4` value. This gives recipient contracts the ability to waste the gas of the `msg.sender` by returning a long revert string (assuming recipient also does not mind reverting the transaction).

Recommendation: There should be little need to pass fine-grained error data when a transfer fails, as in most normal cases it will simply be because the receiving contract doesn't implement the `onERC1155Received()` function; thus, the catch case handling an arbitrary string could likely be removed with little downside. More complicated approaches like limiting the length of the string that is copied from returndata are viable as well.

Response: Fixed in [PR 1243](#).

FPS: Fix verified.

SC.2 ERC-1155 Compatibility: Missing ERC-165 `supportsInterface()` Function

Severity: Informational

Code Location: N/A

Description: The ERC-1155 [standard](#) specifies:

“Smart contracts implementing the ERC-1155 standard MUST implement the ERC-165 `supportsInterface` function and MUST return the constant value `true` if `0xd9b67a26` is passed through the `interfaceID` argument.”

The Hyperdrive implementation does not implement `supportsInterface` at all, meaning it does not comply with the standard.

Recommendation: While there is unlikely to be any significant practical consequence, it is recommended to conform to the standard and implement `supportsInterface` as described.

Response: Fixed in [PR 1243](#).

FPS: Fix verified.

3.2 Code Quality

CQ.1 Comments Do Not Reflect That Full ERC-1155 Support Has Been Implemented

Severity: Informational

Code Location:

<https://github.com/delvtech/hyperdrive/blob/76a91bf1fe8ad32aa59d065a29037d961707c245/contracts/src/internal/HyperdriveMultiToken.sol#L13-L16>

Description: The indicated comments state “This token is mostly ERC1155 compliant; however, we remove on transfer callbacks and safe transfer because of the risk of external calls to untrusted code.”. However, callbacks and safeTransfer have now been implemented, so this comment is no longer accurate.

Recommendation: Update the comments to clarify that full ERC-1155 support has been implemented.

Response: Fixed in [PR 1243](#).

FPS: Fix verified.

4. Notes

This section contains general considerations for interacting with or maintaining the system and various conclusions reached or discoveries made during the course of the assessment. Whereas findings generally represent things for the team to consider changing, notes are more informational and may be helpful to those who intend to interact with the system.

4.1 Review Time Versus System Complexity for Mint and Burn

Prior to this assessment, the reviewer was unfamiliar with the inner workings of the Hyperdrive protocol. It was necessary to understand significant portions of the existing business logic in order to assess the correctness of the mint and burn implementation. The allotted time was not sufficient to obtain full context. While no serious issues were identified with this functionality based on the partial context that was gained, the reviewer is less confident in this result than for adequately-resourced assessments.